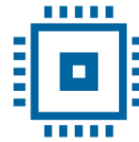




Transilvania
University
of Brasov



Transilvania
University
of Brasov

FACULTY OF ELECTRICAL ENGINEERING
AND COMPUTER SCIENCE

Department of Electronics and Computers

Study programme: Applied Electronics

Verification Project

Interrupt Controller

Table of Contents

1	Digital Design of the Circuit.....	3
1.1	Block Diagram	3
1.2	General Functionality Description of the DUT	3
1.3	DUT Interfaces	4
1.4	List of DUT Functionalities.....	5
1.5	Waveforms	6
1.6	Register Table	7
2	Digital Circuit Project Verification.....	14
2.1	Verification Environment Arhitecture.....	14
2.1.1	APB Generator.....	14
2.1.2	APB Driver	14
2.1.3	APB Monitor.....	15
2.1.4	Output Interface Monitor	15
2.1.5	Transaction for APB Interface.....	15
2.1.6	Transaction for Output Interface.....	15
2.1.7	Scoreboard/Reference Model	15
2.2	List of Elements to be Verified	16
2.3	Functional Coverage Elements.....	17
2.4	Tests.....	18
2.5	Obtained Results.....	18

1 DIGITAL DESIGN OF THE CIRCUIT

Block Diagram

General functionality description of the DUT

DUT interfaces

List of DUT functionalities

Schematic diagram

Waveforms

Register table

1.1 BLOCK DIAGRAM

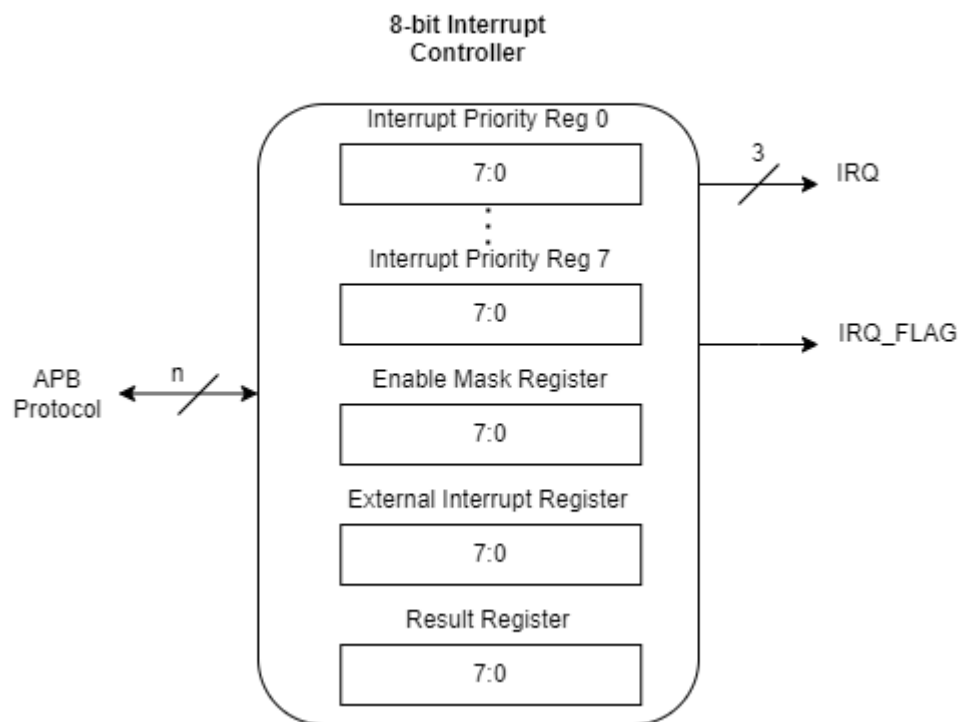


Figure 1. Block diagram of the interrupt controller

1.2 GENERAL FUNCTIONALITY DESCRIPTION OF THE DUT

The DUT receives the APB interface as input and, based on the 10 configuration registers, provides as output the binary code associated with the interrupt channel number that needs to

be serviced. The irq_flag signal indicates the occurrence of an interrupt and validates the output data.

1.3 DUT INTERFACES

Table 1. Signals within the APB data transmission interface

APB Interface			
Signal Name	Bit Width	Direction	Description
psel	1	Input	Selection signal
penable	1	Input	Enable signal. This signal indicates the second cycle of an APB transfer
pwrite	1	Input	Transfer direction. This signal indicates a write access when high and a read access when low
paddr	6	Input	Transfer address
pwwdata	32	Input	Data to be written
prdata	32	Output	Data to be read
pready	1	Output	Ready signal. The slave can use this signal to extend a transfer
pslverr	1	Output	Error signal. This signal indicates a failed transfer

APB transfers are completed in two clock cycles if no wait states are introduced and consist of two phases: setup and access.

- In the setup phase, paddr and pwwdata are set by the Master device.
- psel and pwrite are asserted to indicate that the Slave device is selected and that the access type is a write.
- In the access phase, the Master asserts penable, and the Slave responds with pready to signal that the access has been completed.

Table 2. Signals within the output interface

Output Interface			
Signal Name	Bit Width	Direction	Description
irq	3	Output	Binary code of the active interrupt channel
irq_flag	1	Output	Pulse signal validating the irq output

1.4 LIST OF DUT FUNCTIONALITIES

- In the Interrupt Priority Registers, the value 0 is reserved for unconfigured states.
- In the Enable Mask Register, the value 1 signifies the validation of the respective input.
- The APB register write interface writes 8 bits per call.
- The unmasked interrupt servicing phase begins after all priority registers have been configured and the register representing the input applied to the DUT (EXT_INT) is set.
- The active interrupt servicing ends after serving the last validated input in terms of priority. To start a new servicing cycle, a new value must be written to the EXT_INT register. This clears the internal served-interrupts mask and allows the controller to re-evaluate all enabled, asserted interrupts by priority.
- For the APB interface, the RESULT register is read-only and can only be written to within the DUT.
- The output validation signal (IRQ_FLAG) is pulse-based.
 - DUT Parameters:
 - ADDR_WIDTH: Address length (in bits)
 - DATA_WIDTH: Data length (in bits)
 - IRQ_WIDTH: Interrupt length (in bits)
 - Issues encountered during DUT design:

- Traversing all interrupt priorities using a for loop.
- Solution: Implementing an FSM (Finite State Machine) for each priority register.
- Limitations in DUT operation:
 - pready is generated combinatorially as psel AND penable, meaning it becomes active in the same clock cycle as penable during the access phase.

The DUT implements a three-state FSM: WAIT, PROCESS, and VALIDATE. Within the PROCESS state, a sub-FSM scans all 8 priority registers sequentially (PRI0 through PRI7), followed by a DECIDE state. The DECIDE state allows the final non-blocking assignment from PRI7 to settle before determining whether an eligible interrupt exists. If a valid interrupt is found, the FSM transitions to VALIDATE, which outputs the interrupt code and asserts IRQ_FLAG for one clock cycle. If no eligible interrupt is found, the FSM returns to WAIT.

1.5 WAVEFORMS

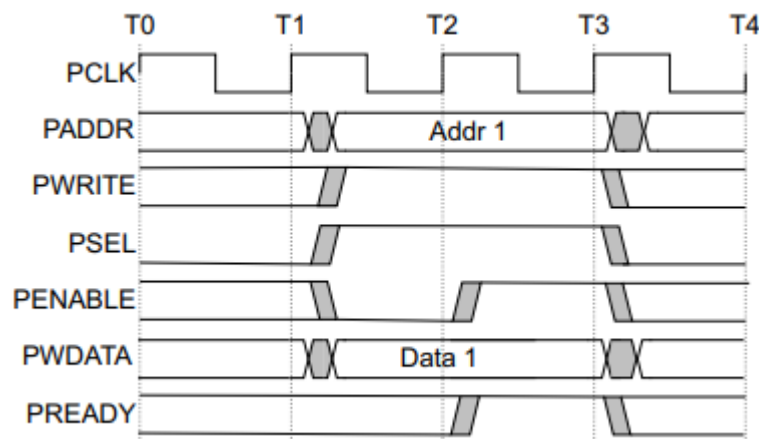


Figure 2. Waveforms for a write transfer using the APB interface

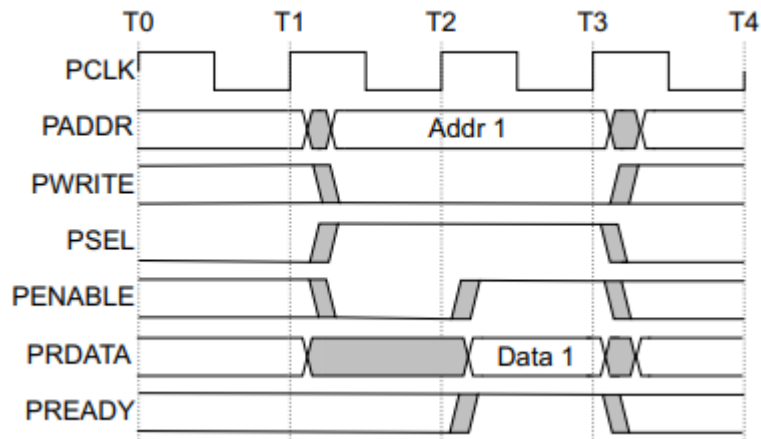


Figure 3. Waveforms for a read transfer using the APB interface

1.6 REGISTER TABLE

The configuration registers are:

1. Interrupt Priority Registers – A set of 8 registers, each 8 bits wide, containing the priority of the external channel on a scale from 1 to 8. The values recorded for the 8 sources cannot be equal.
2. Enable Mask Register – An 8-bit register containing the validation mask for the inputs.
3. Result Register – An 8-bit register that is updated with every change of the outputs.
4. External Interrupt Register – An 8-bit register, one for each input. The values of the 8 interruptible channels received by the DUT through the APB interface are stored in this register.

Writes to the registers are performed per byte. For 8-bit registers, the APB protocol is instantiated only once.

Table 3. Registers Accessible by the DUT

Register Name	Access Address	Register Structure	Register Description
INT_PRI0	0x01	7 6 5 4 3 2 1 0 INT_PRI0	This register contains the priority of input 0 of the module
INT_PRI1	0x02	7 6 5 4 3 2 1 0 INT_PRI1	This register contains the priority of input 1 of the module

INT_PRI2	0x03	7 6 5 4 3 2 1 0 INT_PRI2	This register contains the priority of input 2 of the module
INT_PRI3	0x04	7 6 5 4 3 2 1 0 INT_PRI3	This register contains the priority of input 3 of the module
INT_PRI4	0x05	7 6 5 4 3 2 1 0 INT_PRI4	This register contains the priority of input 4 of the module
INT_PRI5	0x06	7 6 5 4 3 2 1 0 INT_PRI5	This register contains the priority of input 5 of the module
INT_PRI6	0x07	7 6 5 4 3 2 1 0 INT_PRI6	This register contains the priority of input 6 of the module
INT_PRI7	0x08	7 6 5 4 3 2 1 0 INT_PRI7	This register contains the priority of input 7 of the module
EXT_INT	0x09	7 6 5 4 3 2 1 0 EXT_INT7EXT_INT6EXT_INT5EXT_INT4EXT_INT3EXT_INT2EXT_INT1EXT_INT0	This register contains the values of the 8 interrupt inputs
RESULT	0x10	7 6 5 4 3 2 1 0 VALID Reserved INTR_CODE	This register contains the values of the module's two outputs. It is written on every output update.
EN_MASK	0x11	7 6 5 4 3 2 1 0 EN_INT7EN_INT6EN_INT5EN_INT4EN_INT3EN_INT2EN_INT1EN_INT0	This register indicates which inputs are activated and which are ignored (mask functionality register).

Table 4. Interrupt Enable Mask Register

Field	Name	Register Type	Reset Value	Field Description
31:8	RESERVED			
7	EN_INT7	RW	1'b0	1: enabled 0: disabled
6	EN_INT6	RW	1'b0	1: enabled

				0: disabled
5	EN_INT5	RW	1'b0	1: enabled 0: disabled
4	EN_INT4	RW	1'b0	1: enabled 0: disabled
3	EN_INT3	RW	1'b0	1: enabled 0: disabled
2	EN_INT2	RW	1'b0	1: enabled 0: disabled
1	EN_INT1	RW	1'b0	1: enabled 0: disabled
0	EN_INT0	RW	1'b0	1: enabled 0: disabled

Table 5. Interrupt Priority0 Register Configuration

Field	Name	Register Type	Reset Value	Field Description
31:8	REZERVAT			
7:0	INT_PRI0	RW	8'b0000_0000	This register contains the priority value of interrupt code 0 (between 1 and 8)

Table 6. Interrupt Priority1 Register Configuration

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			
7:0	INT_PRI1	RW	8'b0000_0000	This register contains the priority value of interrupt code 1 (between 1 and 8)

Table 7. Interrupt Priority2 Register Configuration

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			
7:0	INT_PRI2	RW	8'b0000_0000	This register contains the priority value of interrupt code 2 (between 1 and 8)

Table 8. Interrupt Priority3 Register Configuration

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			
7:0	INT_PRI3	RW	8'b0000_0000	This register contains the priority value of interrupt code 3 (between 1 and 8)

Table 9. Interrupt Priority4 Register Configuration

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			
7:0	INT_PRI4	RW	8'b0000_0000	This register contains the priority value of interrupt code 4 (between 1 and 8)

Table 10. Interrupt Priority5 Register Configuration

Field	Name	Register	Reset Value	Description
-------	------	----------	-------------	-------------

		Type		
31:8	RESERVED			
7:0	INT_PRI5	RW	8'b0000_0000	This register contains the priority value of interrupt code 5 (between 1 and 8)

Table 11. Interrupt Priority6 Register Configuration

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			
7:0	INT_PRI6	RW	8'b0000_0000	This register contains the priority value of interrupt code 6 (between 1 and 8)

Table 12. Interrupt Priority7 Register Configuration

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			
7:0	INT_PRI7	RW	8'b0000_0000	This register contains the priority value of interrupt code 7 (between 1 and 8)

Table 13. External Interrupt Register

Field	Name	Register Type	Reset Value	Description
-------	------	---------------	-------------	-------------

31:8	RESERVED			
7	EXT_INT7	RW	1'b0	This register contains the value associated with input 7
6	EXT_INT6	RW	1'b0	This register contains the value associated with input 6
5	EXT_INT5	RW	1'b0	This register contains the value associated with input 5
4	EXT_INT4	RW	1'b0	This register contains the value associated with input 4
3	EXT_INT3	RW	1'b0	This register contains the value associated with input 3
2	EXT_INT2	RW	1'b0	This register contains the value associated with input 2
1	EXT_INT1	RW	1'b0	This register contains the value associated with input 1
0	EXT_INT0	RW	1'b0	This register contains the value associated with input 0

Table 14. Result Register (Stores Output Interface Values)

Field	Name	Register Type	Reset Value	Description
31:8	RESERVED			

7	VALID	RO	1'b0	Set to 1'b1 when an interrupt is validated
6:3	RESERVED			
2:0	INTR_CODE	RO	3'b000	Contains the binary code associated with the validated interrupt

2 DIGITAL CIRCUIT PROJECT VERIFICATION

Verification environment architecture

List of elements to be verified

Functional coverage elements

Tests

Obtained results

2.1 VERIFICATION ENVIRONMENT ARCHITECTURE

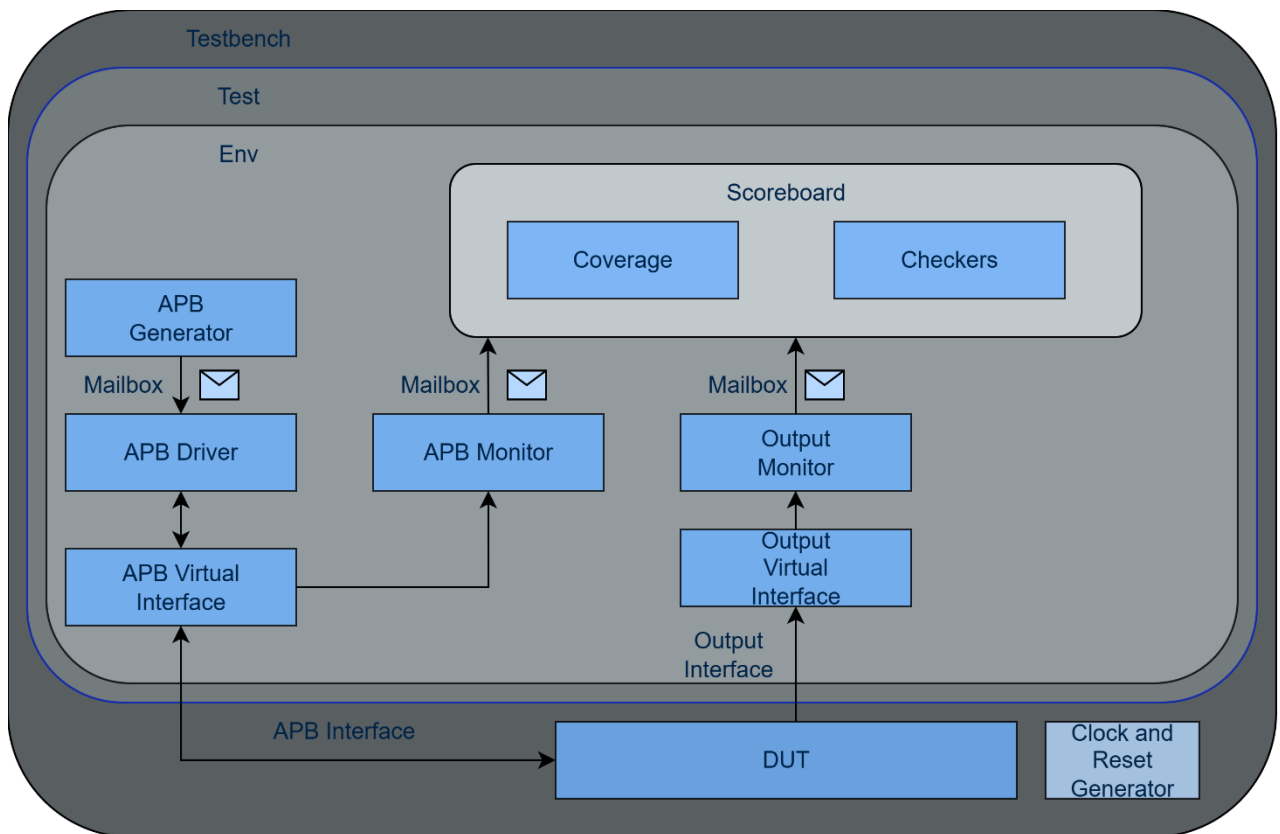


Figure 4. Block diagram of the verification environment

2.1.1 APB Generator

The APB generator randomly generates APB write/read transactions and sends them through a mailbox to the driver.

2.1.2 APB Driver

This component receives transactions from the mailbox and transmits signals to the interface according to the protocol.

2.1.3 APB Monitor

After each transaction sent on the APB interface, the monitor collects signals from the interface and packages them into transactions, which are then sent to the scoreboard. The interface signal that triggers the packaging process is **PSEL**.

2.1.4 Output Interface Monitor

After each transaction sent on the output interface, the monitor checks the signals and packages them into transactions that are later sent to the scoreboard.

Transactions are retrieved from the interface when the **irq_flag** signal is active.

2.1.5 Transaction for APB Interface

The transaction class for the APB interface contains the following fields, which can be randomly generated and sent to the DUT:

- rand bit [5:0] addr
- rand bit [7:0] data
- rand int delay
- rand apb_kind_t kind

2.1.6 Transaction for Output Interface

The transaction class for the output interface contains the following fields, which can be randomized and sent to the DUT:

- rand bit [2:0] irq

2.1.7 Scoreboard/Reference Model

This component contains a prediction model for the DUT's activity. It receives transactions from both the APB interface and the output interface, predicts the expected values of the DUT's registers and outputs, and compares the DUT's actual outputs with the predicted values.

The scoreboard maintains an internal served-interrupts mask to track which interrupts have already been output by the DUT during the current servicing cycle. It uses a priority-based ordering algorithm (`order_interrupts`) that ranks enabled, pending, and unserved interrupts from highest to lowest priority. After each DUT output event, the scoreboard recomputes its prediction to account for the updated served-interrupts state. A pipeline hazard detection mechanism accounts for the DUT's multi-cycle priority scanning window: if a register write occurs within 12 clock cycles of a DUT output, mismatches are reported as warnings rather than errors.

2.2 LIST OF ELEMENTS TO BE VERIFIED

Checker	Description
reg_result_check	Checks whether the result register has the same values as the outputs.
irq_order_check	Checks whether interrupts have been serviced in order of priority.
irq_mask_check	Verifies that the serviced interrupts had their respective bits asserted in the ENABLE_MASK register.

APB Assertions	Description
PSEL_PENABLE	Any rising edge of the psel signal implies a rising edge for the penable signal.
PWDATA_WHILE_WRITE	Specifies that the pwrite write data signal must remain stable (unchanged) during a write operation (when pwrite, psel, and penable are 1).
PADDR_STABLE	Ensures address integrity during transfers on the APB bus by verifying that the address sent by the master to the slave device does not change during the transaction.
PRDATA_WHILE_READ	Verifies that the prdata read data signal does not contain undefined values (X) during a read operation (when pwrite is low, psel is high, and pready is high).
PSEL_DEFINED_STATE	Ensures that the peripheral select signal (psel) does not enter an undefined state (1'bx) or high impedance state (1'bz). It must always have a well-defined logic level (0 or 1).
PSEL_PREADY	Helps verify correct synchronization between the psel (peripheral select) and pready (availability) signals on the APB bus. Ensures that the peripheral completes its current operation before being selected for a new operation.
PSEL_PENABLE_FELL	Verifies that the penable signal is deactivated immediately after the peripheral is deselected.

APB Assertions	Description
PREADY_ONE_PULSE	Ensures that the pready signal remains active for only one clock cycle.
PREADY_DEFINED_STATE	Ensures that when a peripheral is selected and active (psel = 1, penable = 1), the pready signal has a defined value (0 or 1).
PENABLE_AFTER_PREADY	Verifies that the penable signal can only transition from low to high when the psel signal is already asserted, ensuring proper APB protocol sequencing.
PWRITE_DEFINED_STATE	Ensures that the pwrite signal (read/write generated by the master) is always in a defined state, avoiding misinterpretations and potential incorrect operations at the bus level.
PSEL_PENABLE_0	Ensures that when the psel signal rises (beginning the setup phase), the penable signal is not yet asserted, guaranteeing proper APB two-phase transfer sequencing.

Output Validation Assertions	Description
IRQ_FLAG_PULSE	Ensures that the interrupt controller's output validation signal is a pulse signal.

2.3 FUNCTIONAL COVERAGE ELEMENTS

Coverage elements for APB transactions:

Cover Group: apb_transaction_cg	
Cover Point	Description
kind_cov	Both types of transactions (Read, Write) have been recorded.
addr_cov	Transactions have been recorded at all DUT register addresses.
data_cov	Data values between 0 and 255 have been recorded, covering different values in the specified range.

Coverage elements for the output transaction:

Cover Group: out_transaction_cg	
Cover Point	Description
interrupt_cov	All possible interrupt codes (0 to 7) have been recorded on the output interface.

2.4 TESTS

Table 5. List of implemented tests for DUT verification

Test File Name	Test Description
all_intr_en_test.sv	Simulates the configuration of all interrupts, with no masked inputs and all inputs active.
one_intr_en_test.sv	Simulates the configuration of all interrupts, with only one unmasked and active input.
random_test.sv	The registers mentioned above are configured randomly.
write_read_reg_test.sv	Checks the integrity of data written to registers.
addr_invalid_test.sv	Verifies read/write operations at invalid addresses.

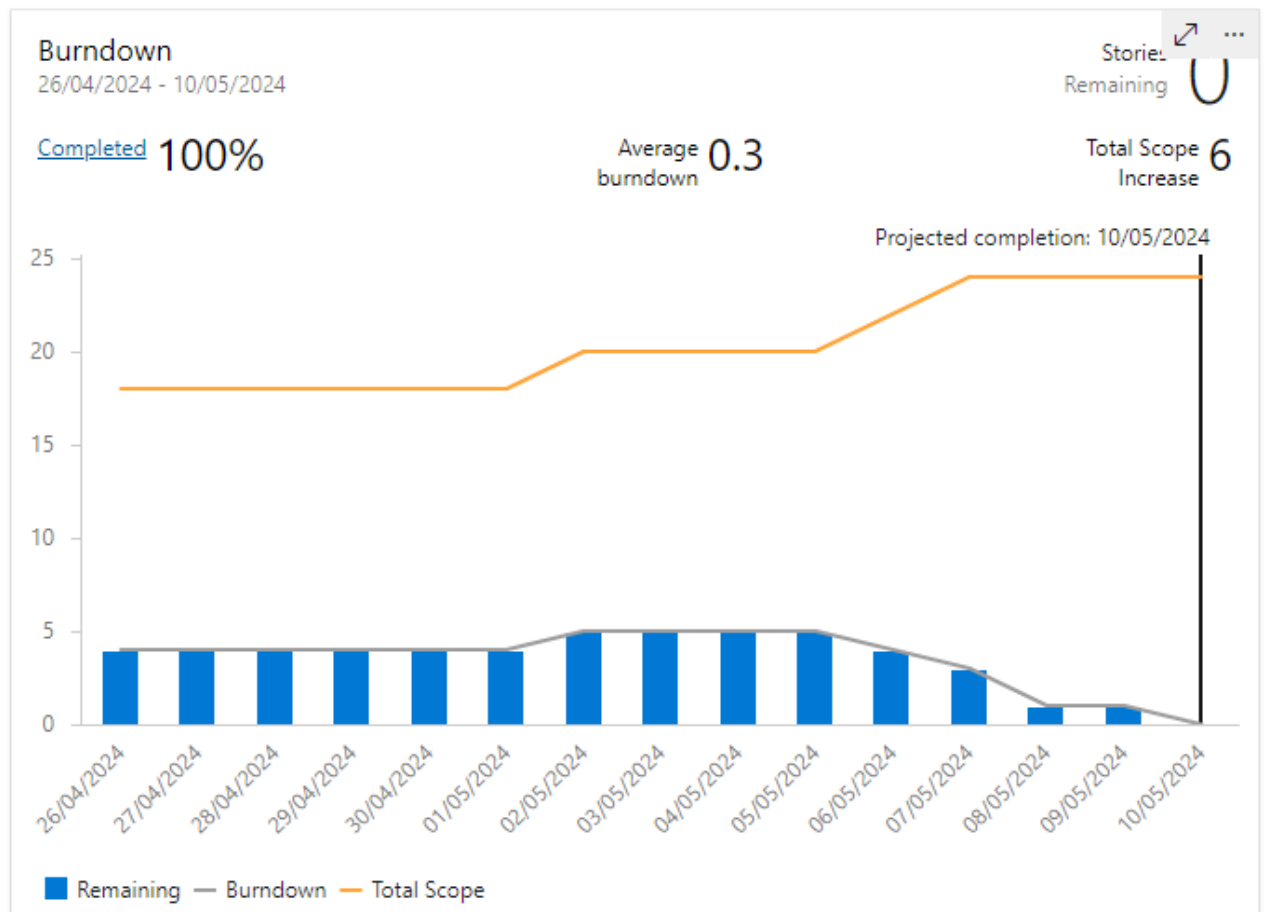
2.5 OBTAINED RESULTS

Each test was executed independently in Vivado behavioral simulation. All tests completed without assertion failures. The functional coverage results are summarized in the table below.

Table 6. Coverage results per test

Test Name	Address Coverage (%)	Data Coverage (%)	Operation Type Coverage (%)	Interrupt Code Coverage (%)
all_intr_en_test.sv	71.43	69.23	50.00	100.00
one_intr_en_test.sv	71.43	61.54	50.00	12.50

random_test.sv	71.43	76.92	50.00	37.50
write_read_reg_test.sv	78.57	100.00	100.00	100.00
addr_invalid_test.sv	7.14	23.08	100.00	0.00



Progress on Task Completion